

Wine Quality Prediction using Machine Learning

Overview

This project aims to predict wine quality based on its chemical properties using Machine Learning classification algorithms. Various features such as acidity, density, alcohol content, and sulphates are used to determine wine quality.

Objectives

- Analyze wine quality data
- Perform data preprocessing and feature analysis
- Visualize relationships between variables
- Train multiple classification models
- Compare model performance

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Machine Learning Models

- Random Forest Classifier
- Stochastic Gradient Descent (SGD) Classifier
- Support Vector Classifier (SVC)

Project Workflow

1. Data Loading
2. Data Cleaning
3. Exploratory Data Analysis
4. Data Visualization
5. Feature Selection
6. Model Training
7. Model Evaluation
8. Performance Comparison

Evaluation Metrics

- Accuracy Score
- Confusion Matrix
- Classification Report

Key Insights

- Chemical properties significantly influence wine quality.
- Alcohol and sulphates show strong relationships with quality ratings.
- Model comparison helps identify the best-performing classifier.

Learning Outcomes

- Classification Algorithms
- Feature Engineering
- Model Evaluation
- Data Visualization
- Predictive Analytics